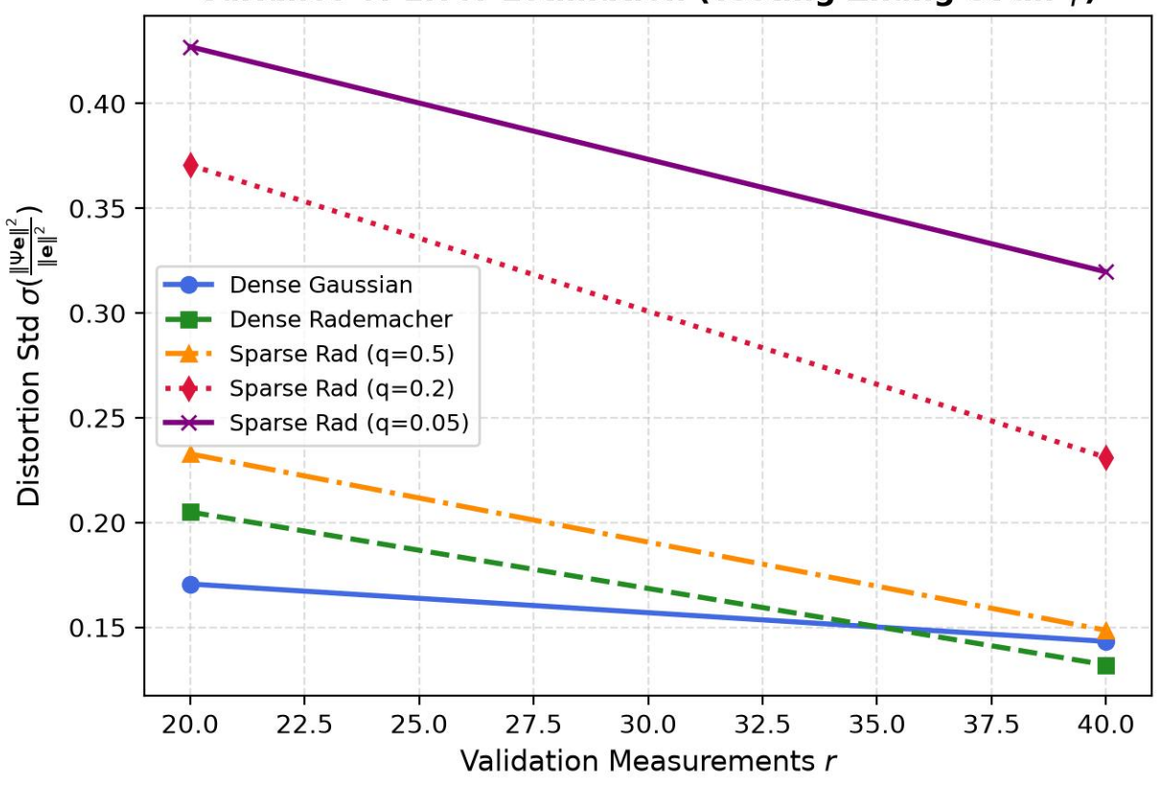
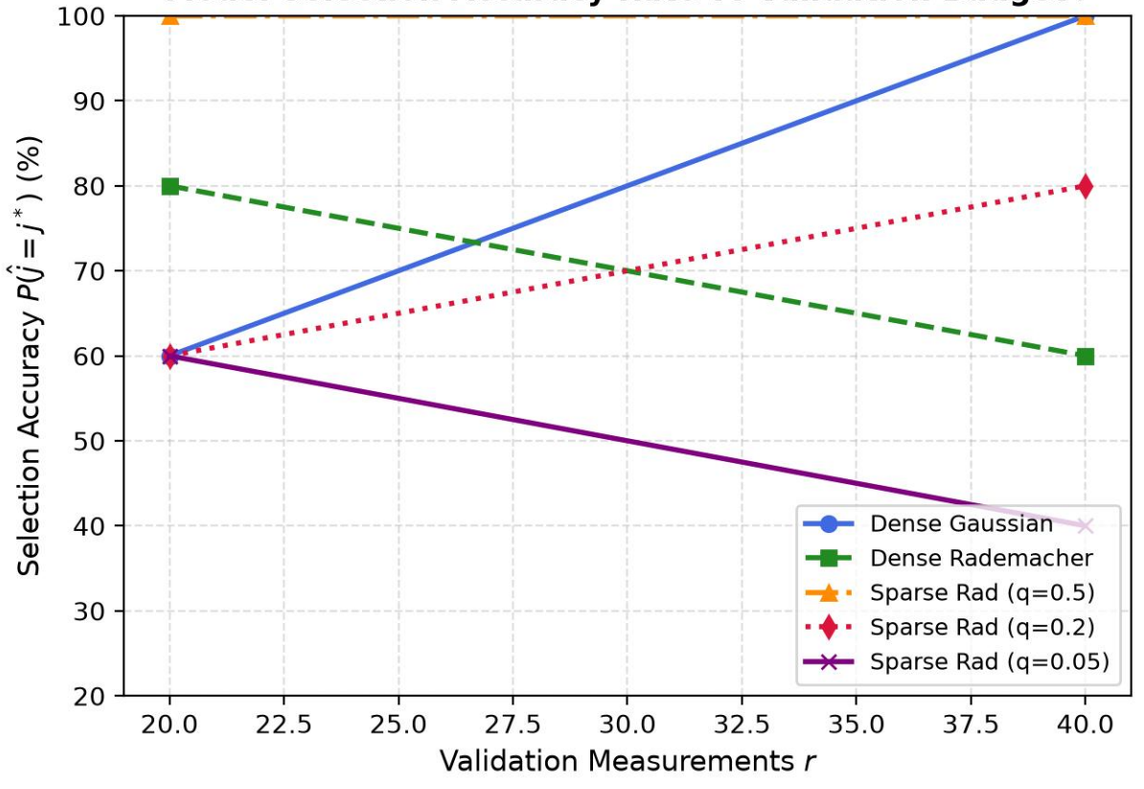
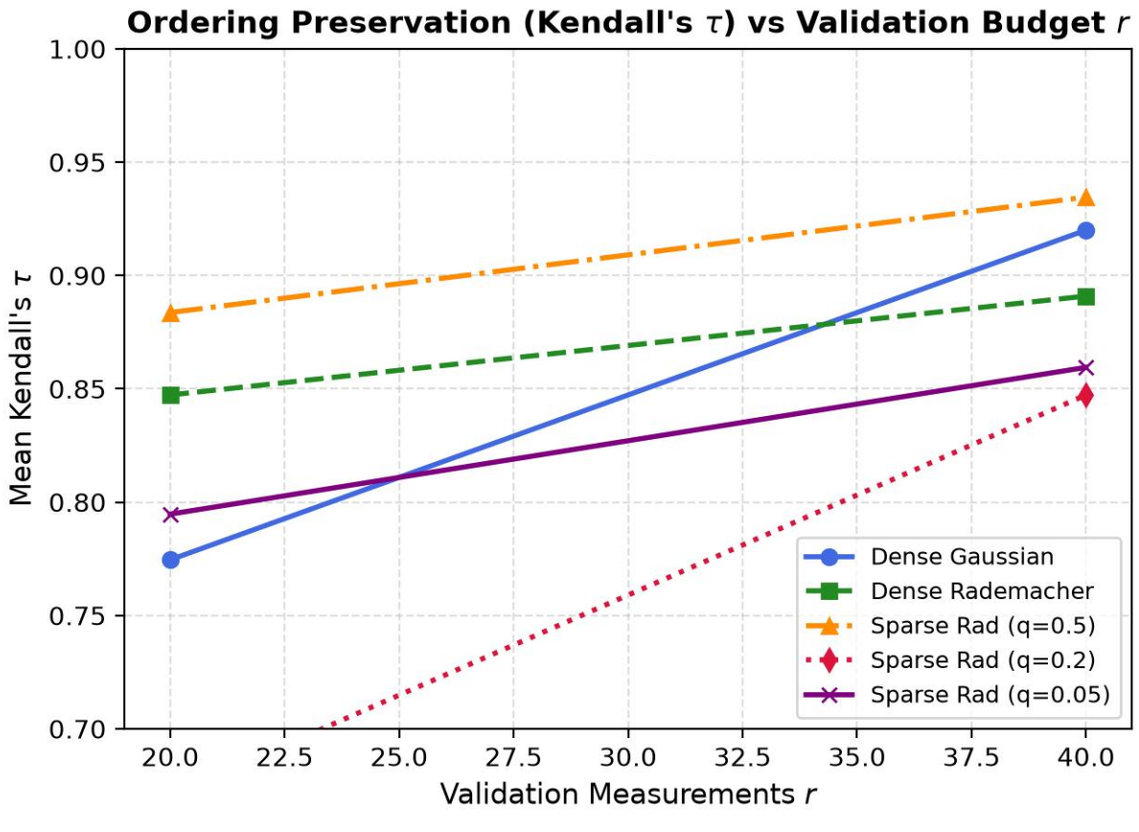
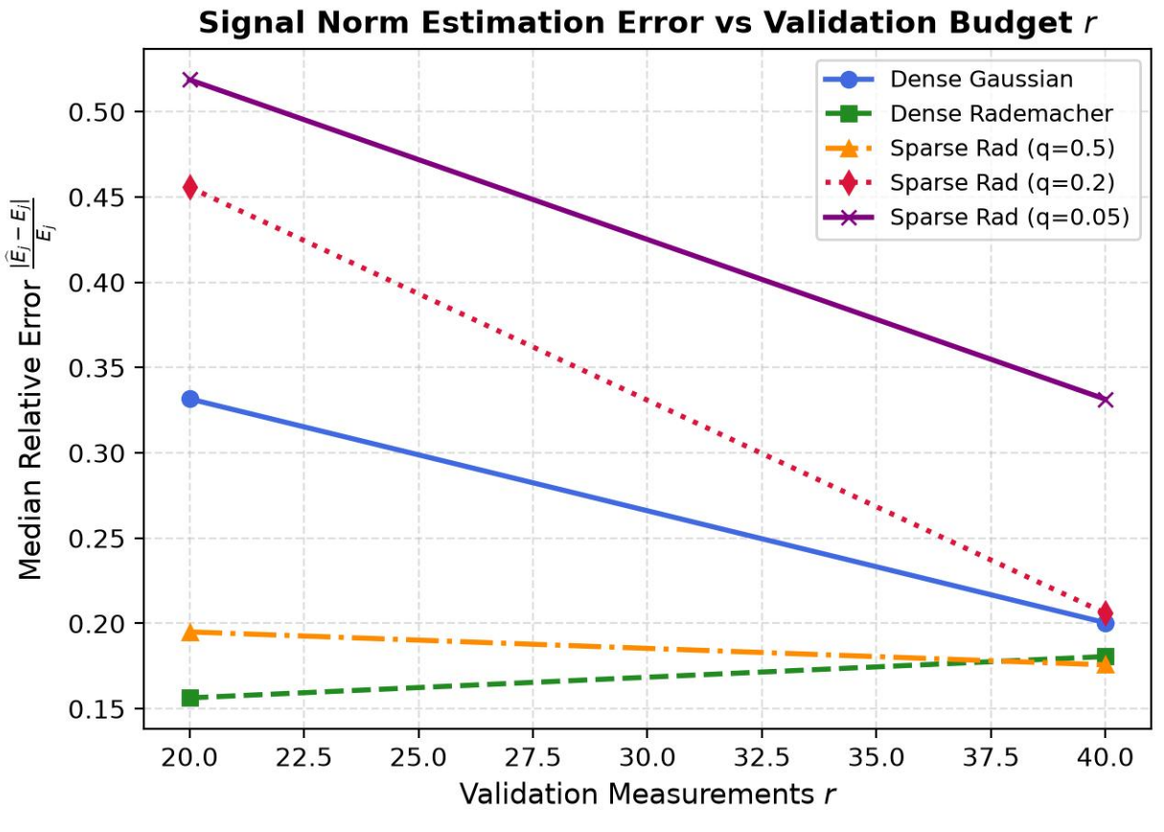

Variance of Error Estimation (Testing Zhang et al. γ)
Distortion Std $\sigma(\frac{\|\Psi e\|^2}{\|e\|^2})$
0.40
0.35
0.30
0.25
0.20
0.15
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5
40.0
Validation Measurements r
Dense Gaussian
Dense Rademacher
Sparse Rad (q=0.5)
Sparse Rad (q=0.2)
Sparse Rad (q=0.05)


Model Selection Accuracy Rate vs Validation Budget r
Selection Accuracy $P(\hat{j}=j^*)$ (%)
100
90
80
70
60
50
40
30
20
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5
40.0
Validation Measurements r
Dense Gaussian
Dense Rademacher
Sparse Rad (q=0.5)
Sparse Rad (q=0.2)
Sparse Rad (q=0.05)


Ordering Preservation (Kendall's τ) vs Validation Budget r
Mean Kendall's τ
1.00
0.95
0.90
0.85
0.80
0.75
0.70
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5
40.0
Validation Measurements r
Dense Gaussian
Dense Rademacher
Sparse Rad (q=0.5)
Sparse Rad (q=0.2)
Sparse Rad (q=0.05)


Signal Norm Estimation Error vs Validation Budget r
Median Relative Error $\frac{|\hat{E}_j - E_j|}{E_j}$
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
20.0
22.5
25.0
27.5
30.0
32.5
35.0
37.5
40.0
Validation Measurements r
Dense Gaussian
Dense Rademacher
Sparse Rad (q=0.5)
Sparse Rad (q=0.2)
Sparse Rad (q=0.05)